MATH 3042- Quiz 5- Solutions

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Full Name:

Time: 30 minutes

Set:

1. The lifetime (in years) of a certain electronic component is exponentially distributed with a mean of 5 years.

a. [2] What is the probability that a component lasts between 4 and 8 years?

$$P(4 \le x \le 8) = \left(1 - e^{-\frac{8}{5}}\right) - \left(1 - e^{-\frac{4}{5}}\right) = 24.74\%$$

b. [2] What is the 90th percentile of component lifetimes?

$$1 - e^{-\frac{t}{5}} = 0.90, \qquad 0.10 = e^{-\frac{t}{5}}, \qquad \ln 0.10 = -\frac{t}{5}, t = 11.51\ years$$

2. You arrive at a metro station at a random time. The train schedule shows that trains depart every 15 minutes (at :00, :15, :30, and :45 past each hour). Since you arrive at a random time within this 15-minute window, your wait time (in minutes) is uniformly distributed between 0 and 15 minutes.

[2] a. What is the probability that you wait 6 minutes or less for the train?

$$P(x \le 6) = \frac{6}{15} = 40\%$$

[2] b. You commute using this train three times this week, arriving at random times each day. What is the probability that you wait more than 12 minutes all three times?

$$P(x > 12) = \frac{3}{15} = 20\%$$
$$P(k = 3) = C(3,3)(0.20)^3(0.80)^0 = (0.20)^3 = 0.008 = 0.8\%$$

[2] c. Using the "unusual values" rule ($|z| > 2$), what wait time would be considered unusually long? Round to one decimal place.

Uniform mean: $\frac{0+15}{2} = 7.5\ minutes$

Uniform var: $\frac{(15-0)^2}{12} = 18.75$

Uniform $\sigma = \sqrt{18.75} = 4.33\ minutes$

$\frac{x-7.5}{4.33} = \pm 2$, $x_1 = -1.16$, $x_2 = 16.16$ both are out of range. There are no unusual wait times.

3. The heights of adult women in a certain population are normally distributed with a mean of **165 cm** and a standard deviation of **7 cm**.

Please write the filled-in R command needed to solve part a.

Let x be the height of a randomly selected adult woman. $X \sim norm(165, 7)$

a. [2] What is the probability that a randomly selected woman is between 160 cm and 175 cm tall?

$P(160 < x < 175) = pnorm(175, 165, 7) - pnorm(160, 165, 7)$

b. [2] What heights mark the middle 90% of this distribution? Given: $qnorm(0.90) = 1.282$, $qnorm(0.95) = 1.645$

$\pm 1.645 = \frac{x - 165}{7}$, $x_{low} = 153.5\ cm$, $x_{high} = 176.5\ cm$

Uniform Probability Distribution

Mean (Expected Value): $\boldsymbol{\mu} = \frac{a+b}{2}$

Variance = $\boldsymbol{\sigma^2} = \frac{(b-a)^2}{12}$

Exponential distribution

$$pdf(x) = \begin{cases} \frac{1}{\beta} e^{-\frac{x}{\beta}} & for\ x > 0 \end{cases}$$

$$cdf(t) = P(X \leq t) = 1 - e^{-\frac{t}{\beta}}$$

Mean and Variance:

$\mu = \beta$ *(mean time between events)*

$\sigma^2 = \beta^2$